

BIOL 250 — Upper Extremity Quick-Reference Notes

Open-note speed reference — built from GR I–VIII answer choices + Martini, Human Anatomy

QUICK-SCAN INDEX

Alphabetical, dense, one line per term — scan this first under time pressure. Full detail is in the sections below.

Term	Quick fact / answer
Abductor pollicis longus	Deep radial n. (= posterior interosseous n.); abducts/extends thumb
Acromioclavicular joint	Plane joint; acromion ↔ clavicle; torn in shoulder separation
Annular + quadrate ligament	Both hold radial head against ulna at proximal radio-ulnar joint
Anterior (flexor) compartment	Arm/forearm front; flexors; forearm splits into superficial + deep layers
Axilla	Armpit — space between upper arm and thoracic wall
Axillary nerve	Posterior cord; deltoid + teres minor; injured in shoulder dislocation
Biceps brachii	Musculocutaneous n. (C5–C6); flexes elbow/shoulder, supinates
Brachialis	Musculocutaneous + radial n.; strongest pure elbow flexor
Brachioradialis	Radial n. (C6–C8); flexes elbow
Brachial plexus order	Roots → Trunks → Divisions → Cords → Branches (Randy Travis Drinks Cold Beer)
Capitate	Largest distal-row carpal; ↔ lunate, trapezoid, hamate, 3rd metacarpal
Carpal mnemonic	Sam Likes To Push The Toy Car Hard (lateral → medial, prox then distal row)
Cervical enlargement	Thicker cord segment where cervical/brachial plexus nerves emerge
Cervical spinal nerve naming	C1–C7 exit ABOVE same-numbered vertebra; C8 exits below C7
Coracobrachialis	Musculocutaneous n. (C5–C7); flexes/adducts arm
Deep branch of radial n.	= Posterior interosseous nerve; supplies deep extensors
Deltoid	Axillary n. (C5–C6); abducts arm (whole muscle)
Dermatome	Skin area innervated by cutaneous sensory branches of one spinal nerve
Dorsal root	Sensory (afferent) root entering spinal cord
Extensor carpi ulnaris	Deep radial n. (= post. interosseous n.); extends/adducts wrist
Extensor digitorum	Radial n.; extends fingers
Fist tendons	Making a fist shows long flexor tendons (PL, FCR, FCU, FDS) at wrist
Flexor carpi radialis	Median n.; flexes + abducts wrist
Flexor carpi ulnaris	Ulnar n.; flexes + adducts wrist
Flexor digitorum profundus	Median n. (lateral half) / ulnar n. (medial half); flexes DIP
Flexor digitorum superficialis	Median n.; flexes PIP + wrist
Glenohumeral joint	Ball-and-socket; most mobile/most-dislocated joint in body
Glenoid labrum	Fibrocartilage rim deepening the glenoid cavity
Hamate	Hook-shaped distal-row carpal, most medial
Humero-ulnar joint	Hinge; trochlea ↔ trochlear notch; main elbow stability
Infraspinatus	Suprascapular n. (C5–C6); laterally rotates arm (rotator cuff)
Intertubercular sulcus	Groove between humeral tubercles carrying long-head biceps tendon
Latissimus dorsi	Thoracodorsal n.; extends/adducts/medially rotates arm
Levator scapulae	C3–C4, dorsal scapular n.; elevates scapula
Long thoracic nerve	C5–C7; serratus anterior; injury → winged scapula
Lunate	Proximal-row carpal next to scaphoid; ↔ radius
Median cubital vein	Connects cephalic + basilic veins; standard blood-draw site
Median nerve	Lateral+medial cords; most anterior forearm flexors, thenar muscles; carpal tunnel
Musculocutaneous nerve	Lateral cord; biceps, brachialis, coracobrachialis
Nerve sheath layers	Endoneurium (axon) < perineurium (fascicle) < epineurium (whole nerve)
Olecranon process	Ulna; point of elbow; fits olecranon fossa on extension

Term	Quick fact / answer
Pectoralis major	Pectoral nerves; flexes/adducts/medially rotates arm
Pectoralis minor	Medial pectoral n.; depresses/protracts shoulder
Pisiform	Pea-shaped proximal-row carpal; FCU tendon attaches here
Posterior (extensor) compartment	Arm/forearm back; extensors; innervated by radial n.
Posterior interosseous nerve	= Deep branch of radial nerve; supplies deep posterior forearm muscles
Pronator quadratus	Median n. (AIN, C8–T1); pronates forearm
Pronator teres	Median n. (C6–C7); 2 heads: medial epicondyle + coronoid process of ulna
Quadratus ligament	Reinforces proximal radio-ulnar joint alongside annular ligament
Radial nerve	Posterior cord; triceps + posterior forearm extensors; wrist drop if injured
Radiocarpal joint	Condylar/ellipsoid wrist joint; flex/extend/ab/adduct
Rhomboid major/minor	Dorsal scapular n. (C5); adduct + downward-rotate scapula
Rotator cuff (SITS)	Supraspinatus, Infraspinatus, Teres minor, Subscapularis
Scaphoid	Most-fractured carpal; lateral-most proximal row
Serratus anterior	Long thoracic n. (C5–C7); protracts scapula; winged scapula if injured
Spinal nerve (thoracic/lumbar/sacral) naming	All exit BELOW their same-numbered vertebra
Subscapularis	Subscapular n. (C5–C6); medially rotates arm (rotator cuff)
Superficial vs deep forearm compartments	Anterior forearm has superficial layer (e.g., FDS) + deep layer (FDP, FPL)
Supinator	Deep radial n. (= post. interosseous n.); supinates forearm
Supraspinatus	Suprascapular n. (C5); initiates abduction (rotator cuff)
Teres major	Lower subscapular n.; extends/medially rotates arm
Teres minor	Axillary n. (C5); laterally rotates/adducts arm (rotator cuff)
Trapezium	Distal-row carpal, thumb side; saddle joint w/ 1st metacarpal
Trapezius	Accessory n. (CN XI); elevates/retracts/depresses/rotates scapula
Trapezoid	Smallest distal-row carpal
Triceps brachii	Radial n. (C6–C8); extends elbow
Triquetrum	Proximal-row carpal, pyramid-shaped; ↔ articular disc/TFCC
Ulnar nerve	Medial cord; FCU, medial FDP, hypothenar + most intrinsic; funny bone
Ventral root	Motor (efferent) root exiting spinal cord
Wrist drop	Sign of radial nerve injury (e.g., Saturday night palsy)
Winged scapula	Sign of long thoracic nerve injury (serratus anterior palsy)

1. Muscles — Action & Innervation

Innervation is bolded/underlined-equivalent via the dedicated column below — this is the column to scan first.

1.1 Scapula movers

Muscle	Origin → Insertion	Innervation	Action	Antagonist
Trapezius	Superior nuchal line, external occipital protuberance, ligamentum nuchae, spinous processes C7–T12 → lateral clavicle, acromion, scapular spine	Accessory n. (CN XI)	Elevate/retract/depress/rotate scapula; extend neck	Serratus anterior (protraction)
Levator scapulae	Transverse processes C1–C4 → superior angle/medial border of scapula	C3–C4, dorsal scapular n.	Elevates scapula	Lower trapezius / pectoralis minor (depression)
Rhomboid major/minor	Minor: spinous	Dorsal scapular n. (C5)	Adduct + downward-rotate	Serratus anterior

	processes C7–T1; Major: spinous processes T2–T5 → medial border of scapula (minor above spine, major below spine)		scapula	(abduct/upward-rotate)
Serratus anterior	Ribs 1–8 (lateral surface) → medial border of scapula (costal surface)	Long thoracic n. (C5–C7)	Protracts shoulder; upward-rotates scapula. Injury → winged scapula	Rhomboids/middle trapezius (retraction)
Pectoralis minor	Ribs 3–5 → coracoid process of scapula	Medial pectoral n. (C8,T1)	Depress/protract shoulder; downward rotation	Trapezius (elevation/upward rotation)
Subclavius	Rib 1 (costochondral junction) → inferior surface of clavicle (subclavian groove)	N. to subclavius (C5–C6)	Depresses/protracts shoulder	Trapezius (elevation)

1.2 Shoulder (glenohumeral) movers — incl. rotator cuff

Muscle	Origin → Insertion	Innervation	Action	Antagonist
Deltoid	Lateral clavicle, acromion, scapular spine → deltoid tuberosity of humerus	Axillary n. (C5–C6)	Whole = abduct; anterior = flex/medial rotate; posterior = extend/lateral rotate	Pectoralis major / latissimus dorsi (adduction)
Supraspinatus (cuff)	Supraspinous fossa of scapula → greater tubercle of humerus (superior facet)	Suprascapular n. (C5)	Initiates abduction	Pectoralis major / latissimus dorsi (adduction)
Infraspinatus (cuff)	Infraspinous fossa of scapula → greater tubercle of humerus (middle facet)	Suprascapular n. (C5–C6)	Lateral rotation	Subscapularis (medial rotation)
Teres minor (cuff)	Lateral border of scapula (upper) → greater tubercle of humerus (inferior facet)	Axillary n. (C5)	Lateral rotation, adduction	Subscapularis / teres major (medial rotation)
Subscapularis (cuff)	Subscapular fossa (anterior scapula) → lesser tubercle of humerus	Subscapular n. (C5–C6)	Medial rotation	Infraspinatus / teres minor (lateral rotation)
Teres major	Inferior angle of scapula → intertubercular groove of humerus (medial lip)	Lower subscapular n. (C5–C6)	Extension, medial rotation	Anterior deltoid / coracobrachialis (flexion)
Latissimus dorsi	Spinous processes T7–L5, thoracolumbar fascia, iliac crest, lower ribs → floor of intertubercular groove of humerus	Thoracodorsal n.	Extension, adduction, medial rotation	Deltoid (flexion) / supraspinatus (abduction)
Pectoralis major	Clavicle (medial half), sternum, costal cartilages 1–6 → intertubercular groove of humerus (lateral lip)	Pectoral nerves	Flexion, adduction, medial rotation	Latissimus dorsi / posterior deltoid (extension)
Coracobrachialis	Coracoid process of scapula → medial shaft of humerus	Musculocutaneous n. (C5–C7)	Flexion, adduction	Triceps brachii long head (extension)

Rotator cuff = SITS: Supraspinatus, Infraspinatus, Teres minor, Subscapularis — all scapula → humerus, all stabilize the glenohumeral joint.

1.3 Elbow / forearm movers

Compartment terminology: the arm and forearm are each divided into an anterior (flexor) compartment and a posterior (extensor) compartment. The forearm's anterior compartment is further split into superficial and deep layers (e.g., FDS = superficial, FDP/FPL = deep).

Muscle	Origin → Insertion	Innervation	Action	Antagonist
Biceps brachii (long+short head)	Long head: supraglenoid tubercle of scapula; Short head: coracoid process → radial tuberosity + bicipital aponeurosis	Musculocutaneous n. (C5–C6)	Flex elbow & shoulder; supinate	Triceps brachii (extend) / pronator teres (pronate)
Brachialis	Anterior distal humerus → ulnar tuberosity/coronoid process of ulna	Musculocutaneous + radial n.	Flex elbow (strongest pure flexor)	Triceps brachii (extend)
Brachioradialis	Lateral supracondylar ridge of humerus → styloid process of radius	Radial n. (C6–C8)	Flex elbow	Triceps brachii (extend)
Triceps brachii (3 heads)	Long head: infraglenoid tubercle of scapula; Lateral + medial heads: posterior humerus → olecranon process of ulna	Radial n. (C6–C8)	Extend elbow; long head also extends/adducts shoulder	Biceps brachii / brachialis (flex)
Anconeus	Lateral epicondyle of humerus → olecranon/posterior ulna	Radial n. (C6–C8)	Extend elbow	Biceps brachii / brachialis (flex)
Pronator teres (2 heads: medial epicondyle of humerus + coronoid process of ulna)	Medial epicondyle of humerus + coronoid process of ulna → lateral surface of radius (mid-shaft)	Median n. (C6–C7)	Pronate forearm + flex elbow	Supinator / biceps brachii (supinate)
Pronator quadratus	Distal anterior ulna → distal anterior radius	Median n. (C8–T1, AIN)	Pronate forearm	Supinator (supinate)
Supinator	Lateral epicondyle of humerus + supinator crest of ulna → proximal lateral radius	Deep radial n. (= posterior interosseous n.) (C6–C8)	Supinate forearm	Pronator teres / quadratus (pronate)

1.4 Wrist / finger movers (forearm)

Muscle	Origin → Insertion	Innervation	Action	Antagonist
Flexor carpi radialis	Medial epicondyle (common flexor origin) → base of 2nd/3rd metacarpals	Median n. (C6–C7)	Flex + abduct wrist	Extensor carpi radialis longus/brevis
Flexor carpi ulnaris	Medial epicondyle + olecranon → pisiform, hamate, base of 5th metacarpal	Ulnar n. (C8–T1)	Flex + adduct wrist	Extensor carpi ulnaris
Palmaris longus	Medial epicondyle → palmar aponeurosis	Median n. (C6–C7)	Flex wrist	Extensor digitorum / extensor carpi radialis
Flexor digitorum superficialis	Medial epicondyle, coronoid process of	Median n.	Flex fingers (PIP) + wrist	Extensor digitorum

	ulna, radius → middle phalanges of fingers 2–5			
Flexor digitorum profundus	Proximal ulna + interosseous membrane → distal phalanges of fingers 2–5	Median n. (lateral half) / Ulnar n. (medial half)	Flex fingers (DIP) + wrist	Extensor digitorum
Flexor pollicis longus	Anterior radius + interosseous membrane → distal phalanx of thumb	Median n. (AIN)	Flex thumb	Extensor pollicis longus/brevis
Extensor carpi radialis longus/brevis	Lateral epicondyle (+ supracondylar ridge for longus) → base of 2nd metacarpal (longus)/3rd metacarpal (brevis)	Radial n. (C6–C7)	Extend + abduct wrist	Flexor carpi radialis
Extensor carpi ulnaris	Lateral epicondyle + olecranon → base of 5th metacarpal	Deep radial n. (= posterior interosseous n.) (C6–C8)	Extend + adduct wrist	Flexor carpi ulnaris
Extensor digitorum	Lateral epicondyle (common extensor origin) → middle/distal phalanges of fingers 2–5 (extensor expansion)	Radial n.	Extend fingers	Flexor digitorum superficialis/profundus
Abductor pollicis longus	Posterior radius/ulna + interosseous membrane → base of 1st metacarpal	Deep radial n. (= posterior interosseous n.)	Abduct/extend thumb	Adductor pollicis
Extensor pollicis brevis/longus	Posterior radius (brevis)/ulna (longus) + interosseous membrane → proximal phalanx of thumb (brevis)/distal phalanx of thumb (longus)	Deep radial n. (= posterior interosseous n.)	Extend thumb	Flexor pollicis longus/brevis

Fast rule: anterior (flexor) forearm = median or ulnar n.; posterior (extensor) forearm = radial n. — almost no exceptions.

Make a fist: multiple long flexor tendons (palmaris longus, FCR, FCU, FDS) become visible/palpable crossing the wrist.

1.5 Intrinsic hand muscles

Muscle group	Origin → Insertion	Innervation	Action	Antagonist
Thenar: abductor pollicis brevis, flexor pollicis brevis, opponens pollicis	Flexor retinaculum, scaphoid, trapezium → proximal phalanx of thumb / 1st metacarpal (opponens)	Median n.	Move thumb (thumb pad)	Adductor pollicis
Adductor pollicis	Capitate + 2nd/3rd metacarpals → proximal phalanx of thumb (medial side)	Ulnar n.	Adducts thumb	Abductor pollicis brevis/longus
Hypothenar: abductor digiti minimi, flexor digiti minimi brevis, opponens digiti minimi	Flexor retinaculum, hamate (hook) → proximal phalanx of little finger / 5th metacarpal (opponens)	Ulnar n.	Move pinky (pinky pad)	— (no classic single antagonist)
Palmaris brevis	Flexor retinaculum/palmar aponeurosis → skin	Ulnar n.	Wrinkles skin of palm	— (cutaneous muscle, no antagonist)

	of medial palm			
Lumbricals (4)	Tendons of flexor digitorum profundus → extensor expansions of fingers 2–5	Median n. (1st–2nd) / Ulnar n. (3rd–4th)	Flex MCP joints, extend IP joints	Flexor digitorum (IP flex) / extensor digitorum (MCP ext)
Dorsal interossei (4)	Adjacent metacarpals (bipennate) → extensor expansions/proximal phalanges (abduct index & ring away from middle, both sides of middle finger)	Ulnar n.	ABduct fingers (DAB)	Palmar interossei (adduct)
Palmar interossei (3–4)	Metacarpals 2, 4, 5 → extensor expansions/proximal phalanges of fingers 2, 4, 5 (toward midline)	Ulnar n.	ADduct fingers (PAD)	Dorsal interossei (abduct)

2. Nerves — Brachial Plexus

C5–T1 → Roots → Trunks (sup/mid/inf) → Divisions (ant/post) → Cords (lateral/posterior/medial) → Terminal nerves. Mnemonic: “Randy Travis Drinks Cold Beer.”

Nerve	Cord	Supplies	Injury clue
Musculocutaneous	Lateral	Biceps brachii, brachialis, coracobrachialis	Weak elbow flexion/supination
Axillary	Posterior	Deltoid, teres minor	Lost abduction after shoulder dislocation
Radial	Posterior	Triceps + all posterior forearm extensors	“Saturday night palsy” → wrist drop
Median	Lateral + Medial	Most anterior forearm flexors, thenar muscles	Carpal tunnel syndrome
Ulnar	Medial	FCU, ulnar half FDP, hypothenar + most intrinsic	“Funny bone” numbness at medial epicondyle

Other named branches: suprascapular n. (suprascapularis, infraspinatus) · thoracodorsal n. (latissimus dorsi) · long thoracic n. (serratus anterior) · dorsal scapular n. (rhomboids, levator scapulae) · subscapular n. (subscapularis, teres major) · axillary n. also → teres minor.

Nerve sheath layer	Location
Endoneurium	Around each axon
Perineurium	Around a fascicle (axon bundle)
Epineurium	Outer sheath of whole nerve

Spinal nerve naming convention (Martini): cervical nerves C1–C7 exit ABOVE their same-numbered vertebra; C8 exits below C7 (above T1). Thoracic, lumbar, and sacral nerves all exit BELOW their same-numbered vertebra.

Dermatome = the area of skin innervated by the cutaneous sensory branches of a single spinal nerve (one pair of dorsal roots).

Cervical enlargement of the spinal cord: the thicker segment of the cord where the cervical/brachial plexus nerves emerge — supplies the upper limb.

3. Bones & Landmarks (one line each)

Term	Quick ID	Articulates with
Clavicle	S-shaped; sternal end ↔ manubrium, acromial end ↔ acromion. Most-fractured bone.	Sternum/manubrium (SC joint); scapula/acromion (AC joint)
Conoid tubercle / costal tuberosity	Clavicle ligament attachment points (acromial / sternal end)	— (ligament attachment, not a joint surface)
Acromion	Scapula process off lateral spine; ↔ clavicle (AC joint); trapezius attaches	Clavicle (acromioclavicular joint)
Coracoid process	Scapula anterior process; origin of short head biceps	— (tendon/ligament attachment, not a joint)

Term	Quick ID	Articulates with
	+ coracobrachialis	surface)
Glenoid cavity	Scapula socket ↔ humeral head = glenohumeral joint	Head of humerus (glenohumeral joint)
Glenoid labrum	Fibrocartilage rim deepening glenoid cavity	— (deepens socket; not itself a separate joint surface)
Supraglenoid tubercle	Origin, long head of biceps brachii	— (tendon origin)
Infraglenoid tubercle	Origin, long head of triceps brachii	— (tendon origin)
Supraspinous / infraspinous fossa	Scapula, above/below the spine — hold supraspinatus / infraspinatus	— (muscle origin sites)
Subscapular fossa	Scapula anterior concave surface — holds subscapularis	— (muscle origin site)
Suprascapular notch	Medial to coracoid base — passage for suprascapular n.	— (nerve passage, not a joint)
Scapula body/neck/borders/angles	Triangle body; superior/medial(vertebral)/lateral(axillary) borders; superior/inferior/lateral angles	— (structural landmarks)
Head of humerus	Proximal/medial; ↔ glenoid cavity	Glenoid cavity of scapula (glenohumeral joint)
Greater tubercle	Humerus lateral; SITS minus subscapularis insert here	— (muscle insertion site)
Lesser tubercle	Humerus anteromedial; subscapularis inserts here	— (muscle insertion site)
Intertubercular sulcus	Between tubercles; carries long-head biceps tendon	— (tendon groove, not a joint)
Anatomical neck	Humerus, at the head — joint capsule border	— (capsule attachment line)
Surgical neck	Humerus, distal to tubercles — common fracture site	— (no joint surface)
Deltoid tuberosity	Humerus lateral shaft — deltoid insertion	— (muscle insertion site)
Radial groove	Humerus posterior shaft — carries radial n. (vulnerable in shaft fx)	— (nerve groove, not a joint)
Capitulum	Humerus distal lateral condyle ↔ radial head	Head of radius (humero-radial joint)
Trochlea	Humerus distal medial condyle ↔ ulna trochlear notch	Trochlear notch of ulna (humero-ulnar joint)
Coronoid / olecranon / radial fossa	Humerus distal anterior/posterior/anterior depressions receiving ulna/radius on flexion-extension	Receive ulna's coronoid process (flexion) / olecranon process (extension); radial fossa receives radial head (flexion)
Medial epicondyle	Humerus; ulnar n. crosses here = “funny bone”	— (muscle origin + ulnar n. passage, not a joint surface)
Lateral epicondyle	Humerus; common origin of forearm extensors	— (muscle origin site)
Olecranon (process)	Ulna; point of elbow; superior lip of trochlear notch	Olecranon fossa of humerus (fits in on extension); part of trochlear notch ↔ humeral trochlea
Coronoid process (ulna)	Inferior lip of trochlear notch	Coronoid fossa of humerus (fits in on flexion); part of trochlear notch ↔ humeral trochlea
Trochlear notch	Ulna ↔ humeral trochlea = humero-ulnar joint	Trochlea of humerus (humero-ulnar joint)
Radial notch (ulna)	Ulna lateral ↔ radial head = proximal radio-ulnar joint	Head of radius (proximal radio-ulnar joint)
Ulnar head / styloid process	Ulna distal end	Ulnar notch of radius + articular disc (distal radio-ulnar joint) — does not directly touch carpals
Head / neck of radius	Proximal radius ↔ capitulum	Capitulum of humerus (humero-radial joint); radial notch of ulna (proximal radio-ulnar joint)
Radial tuberosity	Radius; biceps brachii insertion	— (tendon insertion site)
Radial styloid process	Radius distal lateral — stabilizes wrist	— (ligament attachment, stabilizes wrist laterally)
Ulnar notch of radius	Radius distal ↔ ulnar head = distal radio-ulnar joint	Head of ulna (distal radio-ulnar joint)
Scaphoid	Carpal, proximal row, lateral-most; most-fractured carpal	Radius (radiocarpal joint); lunate, trapezium, trapezoid
Lunate	Carpal, proximal row, next to scaphoid	Radius (radiocarpal joint); scaphoid, triquetrum,

Term	Quick ID	Articulates with
		capitate
Triquetrum	Carpal, proximal row, pyramid shaped	Articular disc/TFCC; lunate, pisiform, hamate
Pisiform	Carpal, proximal row, pea-shaped, most medial	Triquetrum only (sesamoid-like; FCU tendon attaches here)
Trapezium	Carpal, distal row, lateral-most (thumb side)	Scaphoid, trapezoid, 1st metacarpal (saddle joint)
Trapezoid	Carpal, distal row, smallest	Scaphoid, trapezium, capitate, 2nd metacarpal
Capitate	Carpal, distal row, largest	Lunate, trapezoid, hamate, 3rd metacarpal
Hamate	Carpal, distal row, hook-shaped, most medial	Capitate, triquetrum, 4th & 5th metacarpals
Metacarpals I–V	Lateral (thumb) → medial (pinky)	Proximally: carpals (CMC joints). Distally: proximal phalanges (MCP joints)
Phalanges	Thumb = 2 (prox/distal); fingers = 3 (prox/mid/distal)	Adjacent phalanges (IP joints); proximal phalanges also ↔ metacarpals (MCP joints)

Carpal mnemonic (lateral → medial, proximal row then distal): “Sam Likes To Push The Toy Car Hard” = Scaphoid, Lunate, Triquetrum, Pisiform; Trapezium, Trapezoid, Capitate, Hamate.

4. Joints & Ligaments

Joint	Type	Notes
Sternoclavicular	Plane (reinforced)	Only bony link, axial ↔ upper limb; “master joint” of shoulder complex
Acromioclavicular	Plane	Acromion ↔ clavicle; torn in shoulder separation
Glenohumeral (shoulder)	Ball-and-socket, triaxial	Most mobile, most-dislocated joint in body
Humero-ulnar	Hinge	Trochlea ↔ trochlear notch; main elbow stability
Humero-radial	Hinge (part of elbow)	Capitulum ↔ radial head
Proximal / distal radio-ulnar	Pivot	Together = pronation/supination
Radiocarpal (wrist)	Condylar/ellipsoid	Flex/extend/ab/adduct
Intercarpal	Gliding (plane)	Slight sliding/twisting
1st carpometacarpal (thumb)	Saddle	Allows opposition
Metacarpophalangeal (MCP)	Condylar	Flex/extend + some ab/adduct
Interphalangeal (IP)	Hinge	Flex/extend only

Ligament	Connects
Glenohumeral ligaments	Anterior capsule thickenings; limit excess motion
Coracohumeral ligament	Coracoid process → humeral head
Coraco-acromial ligament	Coracoid process → acromion (arches over joint)
Acromioclavicular ligament	Acromion → clavicle
Coracoclavicular ligaments	Clavicle → coracoid process
Transverse humeral ligament	Greater ↔ lesser tubercle; holds biceps long-head tendon in sulcus
Ulnar collateral ligament	Medial epicondyle → coronoid/olecranon (“Tommy John” ligament)
Radial collateral ligament	Lateral epicondyle → annular ligament
Annular ligament + Quadrate ligament	Both hold head of radius against ulna (proximal radio-ulnar joint); annular ligament wraps the radial head, quadrate ligament reinforces the joint capsule below it
Palmar / dorsal radiocarpal ligament	Distal radius → carpal bones, palmar/dorsal side
Ulnar / radial collateral ligament (wrist)	Stabilize wrist medially/laterally

Bursae at the shoulder: subacromial, subdeltoid, subcoracoid, subscapular — cushion tendons crossing the capsule.

5. Vessels

Arterial path: Subclavian → Axillary (after 1st rib) → Brachial (after teres major) → splits at cubital fossa → Radial + Ulnar → superficial/deep palmar arches. First branch of brachial = deep brachial (profunda brachii) a.

Vein	Notes
Cephalic vein	Superficial, lateral side of limb

Vein	Notes
Basilic vein	Superficial, medial side of limb
Median cubital vein	Connects cephalic+basilic at cubital fossa — standard blood-draw site
Brachial vein(s)	Deep, paired, alongside brachial artery
Axillary vein → Subclavian vein	Deep proximal return path

6. Other Real Terms That Show Up as Answer Choices

These aren't UE structures themselves, but they appear as wrong-answer options in UE GRs (cross-sections, posterior trunk views, meninges diagrams). One line each so you can rule them out fast.

6.1 General body-region terms

Term	Means
Dorsum	Back/posterior surface of a body part
Axilla	Armpit — space between upper arm and thoracic wall
Cervicis / cervical	Neck
Cephalon / cephalic	Head
Acromial	Shoulder region (point of shoulder)
Pes	Foot
Hallux	Big toe
Nasus	Nose

6.2 Bone classification (generic, not UE-specific)

Term	Means
Long bone	Shaft + 2 ends (humerus, femur, metacarpals...)
Short bone	Cube-shaped (carpals, tarsals)
Flat bone	Thin plate (scapula body, sternum, skull bones, ribs)
Irregular bone	Complex shape (vertebrae, hip bone)
Sutural (Wormian) bone	Small bones within skull sutures
Sesamoid bone	Forms inside a tendon (patella)

6.3 Trunk / back / neck muscles (distractors, not UE)

Muscle	Innervation	Quick ID
Sternocleidomastoid	Accessory n. (CN XI) + C2–C3	Neck; flexes/rotates head
Platysma	Facial n. (CN VII)	Superficial neck sheet
Omohyoid / sternothyroid	Ansa cervicalis (C1–C3)	Infrahyoid neck muscles
Erector spinae (iliocostalis, longissimus, spinalis)	Dorsal rami	Back extensors, paraspinal
Longissimus thoracis	Dorsal rami	Part of erector spinae
Serratus posterior	Intercostal nerves	Deep back, respiratory accessory
External / internal oblique	Intercostal n. T7–T12 + ilioh./ilioing.	Abdominal wall, opposite fiber directions
Transversus abdominis	Intercostal n. T7–T11 + ilioh./ilioing.	Deepest abdominal wall layer
Psoas major	Lumbar plexus L1–L3	Hip flexor, posterior abdominal wall
Quadratus lumborum	T12–L4 / subcostal n.	Posterior abdominal wall
Rectus sheath / linea alba	—	Connective tissue of anterior abdominal wall, not muscle

6.4 Meninges & spinal cord cross-section terms

Term	Quick ID
Dura mater	Outermost meningeal layer, tough/fibrous

Term	Quick ID
Arachnoid mater	Middle, web-like; borders subarachnoid space (CSF)
Pia mater	Innermost, adheres directly to cord/brain surface
Denticulate ligament	Pia extension anchoring cord laterally to dura
Gray matter	Central H-shape; neuron cell bodies (dorsal/ventral horns)
White matter	Peripheral; myelinated axon tracts
Dorsal root	Sensory (afferent) root entering cord
Ventral root	Motor (efferent) root exiting cord
Dorsal root ganglion	Sensory neuron cell bodies, swelling on dorsal root
Spinal nerve	Union of dorsal + ventral roots
Dorsal ramus	Spinal nerve branch → back muscles/skin
Ventral ramus	Spinal nerve branch → limbs/anterolateral trunk; forms plexuses
Rami communicantes	Connect spinal nerve ↔ sympathetic chain ganglion
Autonomic (sympathetic) ganglion	Sympathetic chain cell bodies, paravertebral

6.5 Other systemic arteries (not UE, but show up near subclavian/aorta questions)

Artery	Quick ID
Ascending aorta / aortic arch	Heart → first major branches of systemic circulation
Brachiocephalic trunk	First branch off aortic arch → R subclavian + R common carotid
Common carotid (L/R)	Neck → internal + external carotid
Internal carotid	→ brain
External carotid	→ face/scalp/neck
Vertebral artery	Subclavian branch → ascends to brainstem; both sides join to form basilar a.
Basilar artery	Formed by R+L vertebral arteries; supplies brainstem/cerebellum
Thyrocervical trunk / internal thoracic artery	Subclavian branches → thyroid/neck / anterior chest wall
Celiac trunk	Abdominal aorta → stomach/liver/spleen
Superior / inferior mesenteric artery	Abdominal aorta → small intestine+colon / distal colon+rectum
Renal artery	Abdominal aorta → kidney

6.6 Lower-limb / pelvic nerves & vessels (not UE — quick recognition only)

Term	Quick ID
Iliohypogastric nerve	Lumbar plexus; abdominal wall/groin
Pudendal nerve	Sacral plexus; perineum
Fibular (peroneal) nerve	Sciatic branch; lower leg
Inferior gluteal nerve	Sacral plexus; gluteus maximus
Lateral femoral cutaneous nerve	Lumbar plexus; sensory, anterior thigh
Great saphenous vein	Longest vein in body; medial leg/thigh
Mastoid process	Skull (temporal bone), behind ear; SCM attachment — not UE